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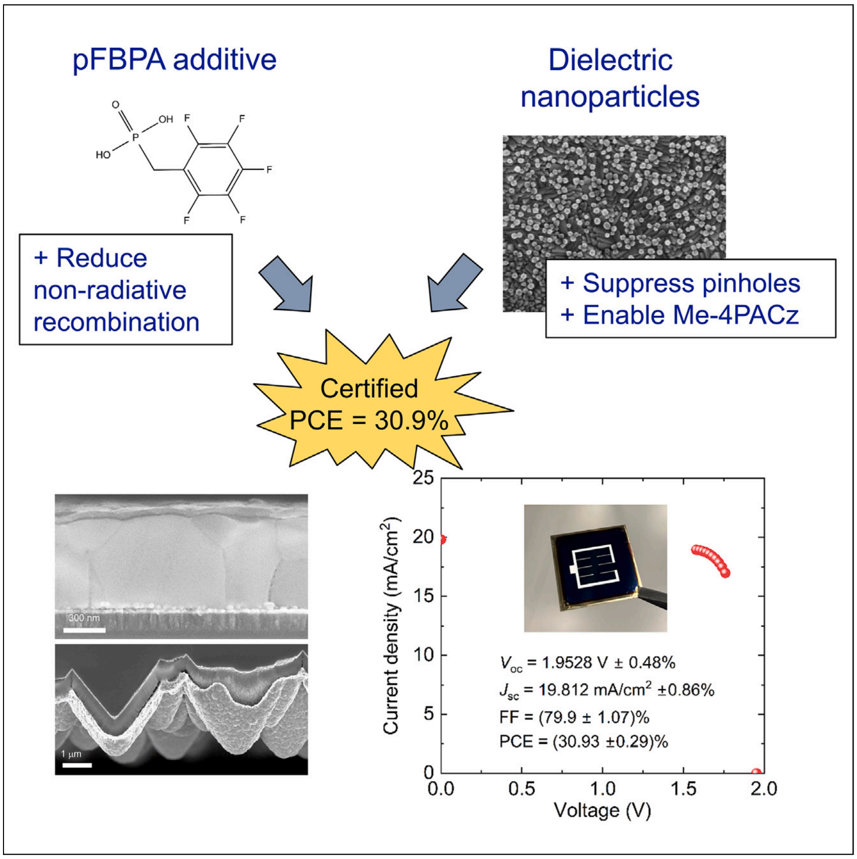
Synergetic substrate and additive engineering for over 30%-efficient perovskite-Si tandem solar cells



Using pFBPA as an additive for solution-processed perovskites significantly suppresses non-radiative recombination. However, it simultaneously deteriorates the film quality, limiting the performance gains. Using dielectric nanoparticles underneath, the film quality can be greatly improved and the gains can be maximized. The nanoparticles also enable the use of Me-4PACz as a hole transport layer, which further improves the performance. Combined with a durable Si cell, a certified power conversion efficiency of 30.9% is achieved for a perovskite-Si tandem device.

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Highlights

pFBPA as an additive to boost the performance of solution-processed perovskites

Silica nanoparticles for improved surface wettability and perovskite film quality

Protective films for the bottom cell, improving performance and reproducibility

Certified perovskite-Si tandem cell with a power conversion efficiency of 30.9%

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Synergetic substrate and additive engineering for over 30%-efficient perovskite-Si tandem solar cells

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SUMMARY

Perovskite-silicon (Si) tandem solar cells are the most prominent contenders to succeed single-junction Si cells that dominate the market today. Yet, to justify the added cost of inserting a perovskite cell on top of Si, these devices should first exhibit sufficiently high power conversion efficiencies (PCEs). Here, we present two key developments with a synergetic effect that boost the PCEs of our tandem devices with front-side flat Si wafers—the use of 2,3,4,5,6-pentafluorobenzylphosphonic acid (pFBPA) in the perovskite precursor ink that suppresses recombination near the perovskite/C_{60} interface and the use of SiO_2 nanoparticles under the perovskite film that suppresses the enhanced number of pinholes and shunts introduced by pFBPA, while also allowing reliable use of Me-4PACz as a hole transport layer. Integrating these developments in an optically and electrically optimized tandem device (e.g., with a durable Si cell), reproducible PCEs of 30 ± 1%, and a certified maximum of 30.9% are achieved.

CONTEXT & SCALE

After decades of research and development, commercial Si solar cells are nearing their fundamental limit in terms of power conversion efficiency. To bypass this limit and further decrease the cost of solar electricity, one promising path is to complement Si with an additional light absorber to use the incoming light more effectively. Metal halide perovskites, with exceptional optoelectronic performance, are one of the prominent candidates for such a combination. Yet, for cost-effective commercialization of this new generation of tandem devices, identification and demonstration of high-performing configurations are vital. Here, we demonstrate a cell design combining additive and substrate engineering that yields consistently high power conversion efficiencies and discuss various design aspects that are crucial for reproducibility and performance.

INTRODUCTION

Commercial photovoltaics has so far been centered around single-junction (SJ) solar cells, with crystalline silicon (Si) being the primary choice of absorber material. However, SJ Si cells have a physical limit to their power conversion efficiency (PCE) that lies slightly above 29%,[1,2] a value not far ahead of the best Si cells of today (27.1%).[3] One way to exceed this limit, and to lower the cost of solar electricity, is to employ an additional absorber with a higher band gap to reduce the thermalization losses and more efficiently use the solar spectrum. Using an absorber with a band gap of around 1.6 to 1.7 eV on Si (with a band gap of 1.1 eV), the PCE limit can be theoretically raised from about 29% to beyond 40%.[4–6] In this respect, perovskite absorbers with tunable band gaps, relatively low fabrication costs, and high optoelectronic performance fit perfectly to Si. Yet, to justify the added cost of inserting a perovskite cell on top of Si, the tandem devices should exhibit both high PCE and operational stability.[7,8] Today, SJ perovskite cells reach up to 26.1% efficiency with small area devices (about 0.1 cm^2), rivaling some of the best-performing Si cells.[3] However, using these cells in a tandem configuration with Si requires modifications (e.g., use of a wide band gap absorber, larger device area [>1 cm^2], change in the direction of incident light) that typically result in deviations from the record-high device performances.[9–11] As a result, perovskite-Si tandem cells are still far from reaching their full potential despite the recent rapid progress.[12–14] For the advancement of the field, identification of reproducible and high-performing cell configurations is vital.

In this work, we present two key developments with a synergetic effect that have been essential in driving the PCEs of our perovskite-Si tandem solar cells (with a spin-coated perovskite film on a front-side flat Si wafer) reliably to 30 ± 1%. The first development includes the use of a phosphonic acid (PA)-based additive (2,3,4,5,6-pentafluorobenzylphosphonic acid, pFBPA) in the perovskite ($Cs_{0.05}[FA_{0.9}MA_{0.1}]_{0.95}Pb[I_{0.8}Br_{0.2}]_3$) precursor ink to eliminate the Pb-related defects and the non-radiative recombination near the perovskite/C_{60} interface, which boosts the open-circuit voltage (V_{oc}) and fill factor (FF) of the devices, notably without the need for an additional passivating layer. The second development, enhancing primarily the FF, includes the use of sparsely coated SiO_2 nanoparticles (NPs) as an interlayer between the perovskite and the hole transport layer (HTL), enabling the reproducible use of the high-performance [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) through improved surface wettability, while also palliating the perovskite film quality deteriorated by the pFBPA. Finally, we give an overview of tandem cells combining these two developments and the rationale behind some of the more general design choices that are aimed to maximize the PCE and reproducibility.

RESULTS

pFBPA for enhanced top cell performance

One of the main problems with the moderately wide band gap (1.65–1.70 eV) perovskite absorbers used in the tandem devices is the large V_{oc} deficit compared with the radiative limit.[15,16] The deficit arises mainly due to bulk and surface defects in the perovskite (e.g., uncoordinated Pb^{2+} and undercoordinated I^- ions), interfacial defects introduced by, for example, the charge transport layers and a non-ideal band alignment throughout the device. In particular with the inverted (*pin*) devices, the non-radiative recombination near the interface of the electron transport layer (ETL, e.g., C_{60}) and the perovskite absorber can severely limit the performance.[17,18] In this respect, use of fullerene derivatives with better energetic alignment than C_{60} and various alternative transport layers (e.g., naphthalene or perylene derivatives) have been demonstrated recently[19–22] Inserting a thin dielectric (e.g., LiF_x, MgF_x, and polystyrene) or a two-dimensional perovskite layer between the C_{60} and the perovskite absorber was also shown to improve particularly the V_{oc} by suppressing the recombination through the defect states introduced by C_{60}.[17,22–25] The Pb-related defects in the perovskite, on the other hand, can be suppressed by using a Lewis-base molecule, either by adding it in the perovskite precursor, in the antisolvent, or using it for a post-deposition surface treatment.[26] The uncoordinated Pb^{2+} ions, which can also transition into metallic Pb^0, can be eliminated by coordination of the functional group of the passivating molecule and the Pb^{2+} or by a change in crystallization dynamics that prevents the formation of Pb defects in the first place.[27,28] To this end, using a molecule with a PA functional group, which can simultaneously passivate the uncoordinated Pb atoms, suppress halide migration, and modify the structural and electronic properties, particularly at the surfaces of the perovskite absorber, is an effective method.[29–31] One such molecule of interest is pFBPA (Figure 1A), which has been used particularly for the modification of metal oxide substrates used with organic electronic devices, in which case it forms a self-assembled monolayer (SAM) on the surface that also improves the adhesion of nonpolar organic molecules.[32–35] However, because of its hydrophobic nature due to the fluorinated benzyl group, pFBPA is not suited for use as a substrate modifier for perovskite devices without additional modifications that enhance surface wettability for solution processing. Recently, the molecule was used to modify the top surface of $FAPbI_3$ films by dip coating for improved performance and humidity resistance with SJ *nip* devices and with the hybrid evaporated and spin-coated $Cs_{0.18}FA_{0.82}Pb(I_xBr_{1-x})_3$ films (by adding it into the organohalide precursor ink) used in the

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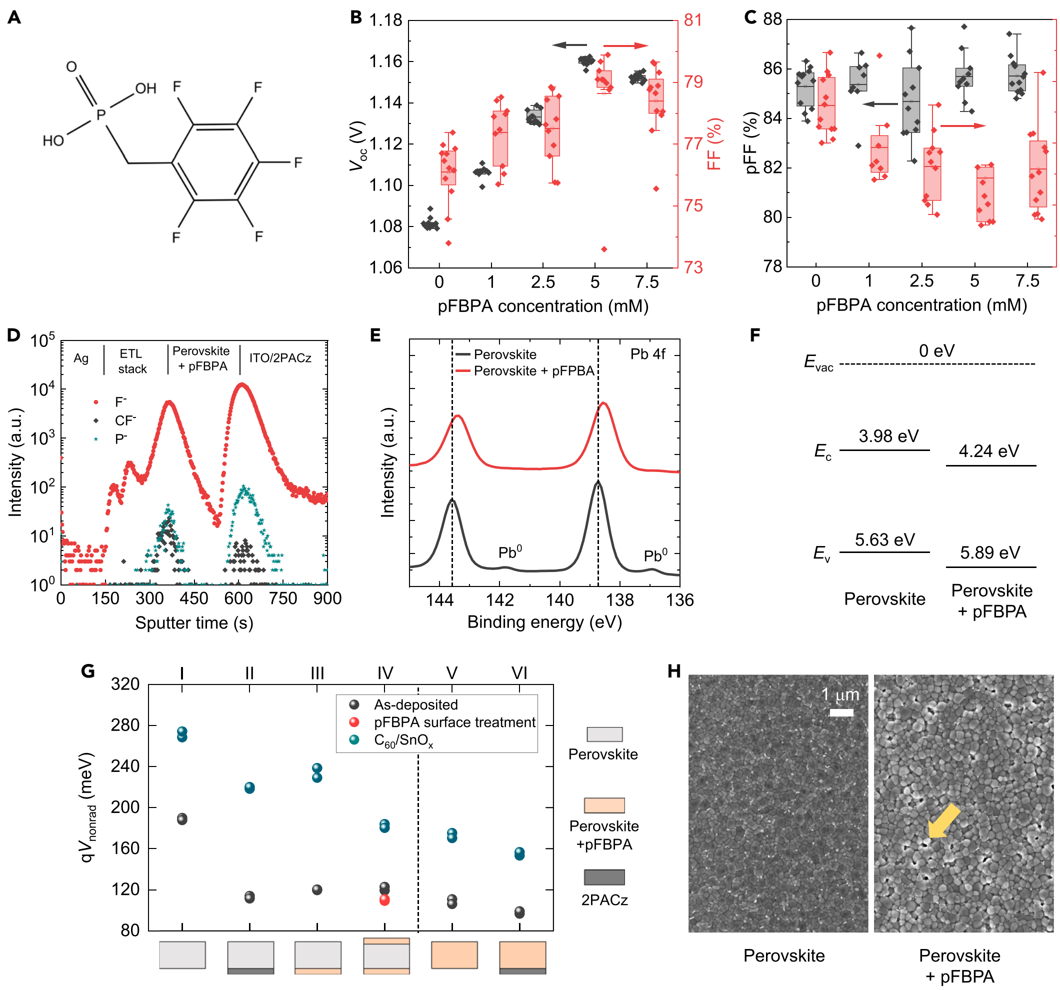

Figure 1. pFBPA as an additive

(A) Chemical structure of 2,3,4,5,6-pentafluorobenzylphosphonic acid (pFBPA).

(B and C) (B) Open circuit voltage (V_{oc}), fill factor (FF), (C) pseudo-FF (pFF) and the series resistance near the maximum power point (R_s) for single-junction 0.1 cm^2 perovskite solar cells on glass/ITO/2PACz substrates with varying pFBPA concentration in the perovskite precursor ink.

(D) Depth profiles of F^-, CF^-, and P^- in finished cells with 5 mM pFBPA in the precursor, as measured by secondary ion mass spectroscopy (SIMS).

(E and F) (E) X-ray photoelectron spectroscopy (XPS) spectra of the Pb 4f core level and (F) the energy levels of the vacuum (E_{vac}), conduction (E_c), and valence band (E_v) of bare perovskite films, with and without pFBPA.

(G) Quasi-Fermi-level splitting loss due to non-radiative recombination ($qV_{non\text{-}rad}$) in half-finished perovskite cells on glass/ITO substrates extracted by steady-state photoluminescence quantum yield measurements.

(H) Scanning electron microscopy images of perovskite absorbers, with and without pFBPA, on glass/ITO/2PACz substrates viewed from the top side. The yellow arrow indicates the location of a pinhole as an example.

record-high-efficiency *pin* tandem devices with fully textured Si wafers.[13,36] Here, we add pFBPA into the perovskite precursor ink that we use in an anti-solvent-assisted spin-coating and crystallization process and improve both the V_{oc} and FF without employing a post-deposition passivation treatment or an additional passivating layer. The amount of incorporated pFBPA into the precursor plays a crucial role on the performance and the response of the device to light exposure and, in the following, we investigate the effect of pFBPA using half-finished and finished (i.e., metallized) SJ cells with various configurations.

In terms of device performance, the pFBPA addition affects the initial response of the cells to light soaking (<3 min exposure), particularly under open-circuit conditions. Here, we do not attempt to uncover the possibly various physical mechanisms (e.g., light and electric-field-induced ion generation and migration) behind these observations, which are not uncommon for perovskite devices.[37,38] Yet, we adjust our measurement methodology to account for this difference in light soaking behavior and precondition the cells to maximize the PCE for each configuration to compare them in their highest-performing states (Figures S1 and S2). Overall, we observe that both the FF and V_{oc} increase with low concentrations of pFBPA ($\leq$5 mM) added in the precursor (Figure 1B). The increase in FF is driven by the reduction in the series resistance at maximum power point (MPP) (R_s), while the pseudo-FF (pFF), correlated with shunting and the recombination rate at both the MPP and V_{oc} conditions, remains roughly the same despite the increase in V_{oc} (Figure 1C). This leads to an increase in the V_{oc} × (p)FF product from 0.83 (0.92 V) to 0.92 V (0.99 V), with a maximum at a concentration of around 5 mM, which is typically the concentration we use in our devices. With excessive pFBPA use (>5 mM), both the V_{oc} and FF decrease, with the latter primarily driven by the increase in R_s.

To gain more insight on the role of pFBPA on device performance, we analyzed finished and half-finished cells at various stages of fabrication. Secondary ion mass spectroscopy (SIMS) measurements of finished cells show that the pFBPA accumulates primarily at the top and bottom surfaces of the films and is present in relatively lower concentrations within the bulk (Figure 1D), which can be associated with the extrusion of the pFBPA molecule toward the surfaces during film formation.[39] X-ray photoelectron spectroscopy (XPS) measurements of bare perovskite films show that the formation of Pb^0 is suppressed at the top surface by using pFBPA (Figure 1E), in alignment with previous works.[36,40] The shift of the Pb 4f peaks to lower binding energies indicates a Lewis acid-base reaction between the pFBPA and Pb^{2+} and the passivation of the Pb^{2+} ions, which appears to be due to the formation of P–O–Pb bonds (Figure S4) and to the likely interaction of the lone electron pairs of P=O and the Pb^{2+} ions.[41] Notably, the suppression of Pb^0 is also accompanied by a reduced amount of PbI_2 in the films (Figure S5), which can reversibly transition into Pb^0.[28,42] Moreover, the valence band maximum (VBM) of the perovskite films is increased by 0.26 eV upon pFBPA addition relative to the vacuum level (Figure 1F). Because the band gap of the perovskite films appears to remain unchanged with pFBPA addition (Figure S6), it can be inferred that the conduction band minimum (CBM) also shifts with the VBM. The precise influence of these changes on the electrical conditions at ETL and HTL interfaces (e.g., on the charge transport across the interfaces) requires in-depth work ideally combining various characterization methods and numerical modeling, which are out of the scope of this work.[43] However, the changes in R_s in combination with those of the CBM and VBM of the perovskite absorber (Figures 1B and 1C) hint toward simultaneously changing band offsets of the transport layers and the perovskite due to the pFBPA use, which can be interpreted to a degree based on the data available on these material systems in literature (Figure S30). For instance, the shift of 0.26 eV can reduce the VBM offset between the perovskite (i.e., from 5.63 to 5.89 eV) and the indium tin oxide (ITO)/2PACz (i.e., between 5.9 to 6.0 eV), which can contribute to the reduction in R_s with pFBPA use. The fact that R_s reaches a minimum at a lower pFBPA concentration when used with Me-4PACz (instead of 2PACz) as the HTL (Figure S3) might be related to the already close VBMs of the Me-4PACz (i.e., 5.66 to 5.80 eV) and the perovskite without the pFBPA use (i.e., 5.63 eV). However, we note that the band structure at the buried HTL interface, which also appears to accommodate more pFBPA molecules than at the surface (Figure 3D), can be different than at the top surface of the perovskite and requires more advanced techniques for its further and more precise analysis.[44] As for the C_{60}/perovskite interface, interpretation of the band alignment remains challenging due to the relatively wide range of CBM data in the data available in literature (i.e., between 4 to 4.5 eV). The pFPBA use can be beneficial for the electron transport if the CBM of the C_{60} lies toward 4.5 eV, in which case the shift of perovskite CBM from 3.98 to 4.24 eV reduces the band offset between the two layers and can contribute to a reduction in R_s. However, the transport can also be hindered by the pFBPA use at this interface if the CBM of the C_{60} lies toward 4 eV, which is already very close to the CBM of the perovskite without pFBPA use. Nevertheless, even if there is a negative influence of pFBPA on this interface, it appears to remain mild compared with the benefit in other parts of the device, as a net gain in device R_s is obtained with the pFBPA use (Figure 1C). Finally, we note that it can be possible to process and optimize the band alignment at the top and bottom sides individually to boost the performance even further (e.g., by spin coating the pFBPA). Yet, such an approach increases the fabrication complexity and number of deposition steps, which we intend to keep minimized by using pFBPA as an additive.

To gain a better insight on the V_{oc} improvement by the pFBPA use and to decouple the non-radiative recombination losses occurring at the perovskite bulk and the ETL and HTL interfaces, we also analyzed perovskite films on glass/ITO and glass/ITO/SAM substrates by steady-state photoluminescence quantum yield measurements (Figure 1G). Although the former material configuration is not suited for high-performance device operation, it is useful for demonstrating the influence of pFBPA at the bottom surface. Without the pFBPA, the quasi-Fermi-level splitting loss due to non-radiative recombination at 1-sun illumination ($qV_{non-rad}$) of a perovskite film on a bare ITO substrate (I) corresponds to 190 meV in the as-deposited state before the subsequent ETL depositions. When pFBPA is added in the precursor, on the other hand, the $qV_{non-rad}$ decreases to roughly 110 meV (V), which is slightly larger than that of a pFBPA-containing perovskite coated on ITO/2PACz at 100 meV (VI). This indicates that the pFBPA accumulated at the bottom surface provides significant passivation in the absence of another SAM on the ITO. However, the charge transport is very poor at this interface despite the low $qV_{non-rad}$ (Figure S7). Notably, it was recently also demonstrated by Zheng et al. that adding Me-4PACz into the perovskite precursor can enable both high quality charge transport and passivation on bare ITO,[45] and thus the concept is indeed intriguing, yet one we do not explore further in this work for pFBPA. On glass/ITO/2PACz substrates, the addition of pFBPA in the precursor results in a decrease in the $qV_{non-rad}$ by roughly 10 meV (II vs. VI). However, in this configuration, the gain by the pFBPA use appears to be mainly due to the difference in the passivation of the top side and possibly at domain boundaries and not due to an improvement at the bottom side. For instance, when pFBPA is spin coated on a perovskite film without the pFBPA in the precursor, the $qV_{non-rad}$ is similar to a film with the pFBPA in the precursor (IV vs. V). Similarly, when IPA is spin coated to wash away the pFBPA at the top surface, the $qV_{non-rad}$ is comparable to that of a sample without the pFBPA in the precursor (Figure S8). Thus, a clear benefit of having pFBPA on the bottom side, where a 2PACz layer is already present, is not observed. The most distinct benefit of using pFBPA appears following the ETL stack depositions (C_{60} and SnO_x), in which case the $qV_{non-rad}$ difference between the samples with and without a pFBPA layer at the top side becomes highly pronounced (roughly 60 meV) (II vs. VI and III vs. IV). This shows that the high non-radiative recombination rate near the perovskite/C_{60} interface is suppressed by the pFBPA. Although this might be due to the suppressed number of Pb-related defects and the modified band structure near the perovskite/C_{60} interface (Figure 1F), it is possible that the defect profile of the C_{60} layer is modified when deposited on the pFBPA-modified perovskite surface, similar to the case previously by Menzel et al. for a C_{60} layer deposited on LiF_x.[46]

In terms of the structural properties of the perovskite films, the thickness is reduced (i.e., by about 10%) and the domain size is increased slightly with the use of pFBPA (Figure S5). Yet, more crucially, a significant number of pinholes are introduced, possibly due to the hydrophobic fluorinated termination group of pFBPA and its extrusion toward the surfaces that affects film formation on a nanoscopic scale (Figure 1H). This also appears to be responsible for the near-constant pFFs (roughly 85%–86%), despite the increase in V_{oc}, with the pFBPA addition, which limits the gain that can be achieved with this additive in a device. Fortunately, the deteriorated film quality and the pFF is recovered to a large degree by using SiO_2-NPs, which also enables the use of Me-4PACz as an HTL, as discussed in the following section.

SiO_2 NPs for enhanced wettability and perovskite film quality

The ITO/Me-4PACz stack enables some of the best performances for *pin* devices with moderately wide band gap perovskite absorbers.[24,47] Yet, the wettability of Me-4PACz by the perovskite solution is quite poor due to the nonpolar methyl termination group of Me-4PACz, resulting in incomplete coverage of the substrate by the perovskite film and a low device yield.[48,49] Recently, various solutions have been proposed to this problem. Improved wettability was demonstrated by texturing the Si surface,[48] by combining Me-4PACz with 1,6-hexylenediphosphonic acid,[49] and by incorporating N-methyl-2-pyrrolidone (NMP) in the perovskite precursor solvent system as an addition to dimethyl formamide (DMF) and dimethyl sulfoxide (DMSO).[50] The use of Al_2O_3 nanoplate islands was demonstrated to improve the PCE of inverted perovskite devices, in which case a partial contact mechanism, similar to that used in passivated emitter and rear contact (PERC) Si cells, was described.[51] Similarly, wettability improving, sparsely coated spherical Al_2O_3 NPs were used in combination with Me-4PACz for small-area *pin* SJ devices but without emphasis or insights on their operation.[52] The use of such sparse Al_2O_3 and SiO_2 NPs as universal wettability agents for perovskite solar cells was, in fact, first demonstrated by Schultes et al. and shortly thereafter by You et al., in which cases they were used with the hydrophobic HTLs (e.g., poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine [PTAA]) and fullerene-based ETLs (e.g., C_{60}-SAM), respectively.[53,54] Yet, the first examples of their use as substrate modifiers on GaN light-emitting diodes (LEDs) and copper indium gallium selenide (CIGS) solar cells predate the aforementioned examples in perovskite research.[55–57] In a broader context, works on the use of partial-contact architectures in various types of thin-film solar cells can also be found in literature, without necessarily focusing on improved surface wettability.[58–60] Classifying the relatively thick and dense dielectric mesoporous layers as partial contact architectures, the list of literature on the subject and its history in perovskite research can be further extended.[61,62]

In our devices, we spin coat sparsely distributed SiO_2-NPs (spheres with an average diameter of roughly 25 nm) on ITO/SAM substrates (Figure 2A), which results in a partial contact between the HTL and the perovskite absorber. The NPs serve two main purposes—enabling the use of Me-4PACz due to improved surface wettability by the hydrophilic SiO_2-NPs and improving the film quality deteriorated by the pFBPA. Figures 2B–2D illustrates a comparison of SJ devices fabricated with 2PACz and Me-4PACz coated with varying dispersion concentrations of SiO_2-NPs (Figures S9 and S10). Comparing 2PACz (without the NPs) with Me-4PACz/SiO_2-NPs (e.g., with 0.1 or 0.2 wt %), a distinct improvement in FF (from $2\%_{abs}$ to $5\%_{abs}$), accompanied by a minor change in V_{oc} (<5 mV), is observed reproducibly

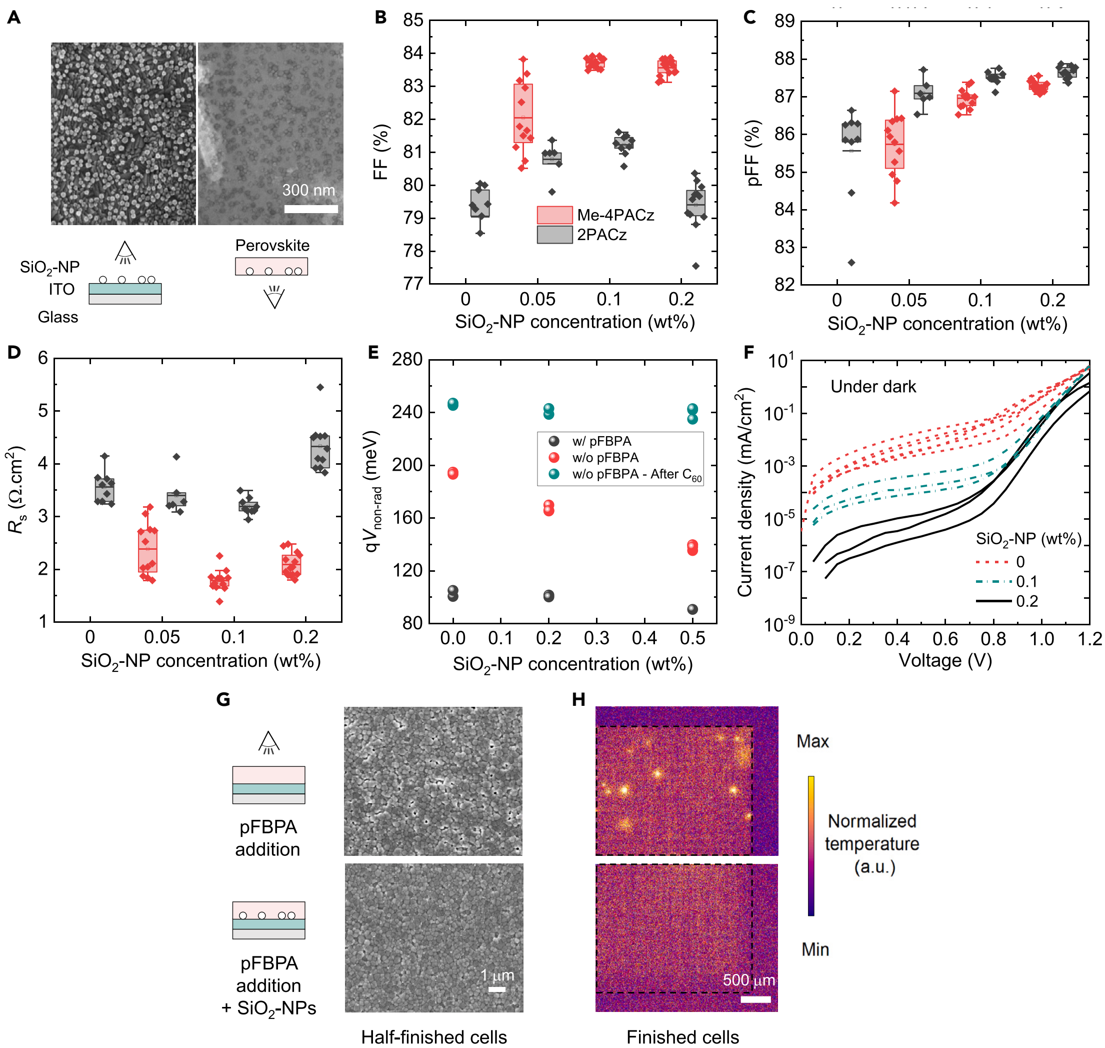

Figure 2. SiO_2 nanoparticles for enhanced surface wettability and film quality

(A) Scanning electron microscope (SEM) images of SiO_2-NPs on a glass/ITO substrate before perovskite coating and SiO_2-NPs embedded under a perovskite film, imaged by peeling off the perovskite film after the coating. The eye symbols represent the location of the detector imaging the films.

(B–D) (B) Fill factor, (C) pseudo-FF (pFF), and (D) series resistance near maximum power point (R_s) of 0.1 cm^2 single-junction perovskite solar cells with varying concentrations of SiO_2-NPs on 2PACz and Me-4PACz hole transport layers.

(E) Quasi-Fermi-level splitting losses due to non-radiative recombination ($qV_{non\text{-}rad}$) of perovskite absorbers with (w/) and without (w/o) pFBPA in the precursor, deposited on glass/ITO substrates (without 2PACz) coated with SiO_2-NP dispersions with varying concentrations.

(F) Current-voltage characteristics extracted under dark conditions for 0.1 cm^2 single-junction perovskite solar cells on glass/ITO/2PACz substrates with varying concentrations of SiO_2-NPs.

(G and H) (G) SEM images of half-finished cells (perovskite films on glass/ITO/2PACz substrates) and (H) lock-in thermography images of finished cells recorded at a forward bias of 1 V under dark conditions, with and without the SiO_2-NPs (0.2 wt %). The black dashes in thermograph images mark the border of the Ag contact.

(Figures S11 and S12). To gain more insight into the operation of this contact architecture and to decouple the individual contributions of the SiO_2-NPs and Me-4PACz on cell performance, it is instructive to analyze both finished and half-finished devices with varying NP surface coverage fractions, as discussed in the following.

Figure 2E shows the $qV_{non\text{-}rad}$ in half-finished devices as a function of SiO_2-NP concentration on glass/ITO substrates, with varying degrees of interface passivation with the perovskite absorber. For a poor bottom interface (i.e., ITO/perovskite without a SAM on ITO or pFBPA in the perovskite), the reduction in q$V_{non\text{-}rad}$ by using SiO_2-NPs is evident even for moderate concentrations (i.e., 0.2 wt %). However, for an interface with already high passivation quality (e.g., with ITO/2PACz or pFBPA), the reduction in q$V_{non\text{-}rad}$ is not substantial (<10 mV) until high NP concentrations, which are detrimental for device operation, are used (e.g., >0.2 wt %). Following the C_{60} deposition, the variation in q$V_{non\text{-}rad}$ with the SiO_2-NP concentration becomes less clear because the high recombination rate near the C_{60} layer dominates the cumulative recombination rate in the samples. With completed SJ devices on glass/ITO/2PACz substrates, on the other hand, distinct variations in device performance, particularly in terms of pFF and R_s (Figures 2C and 2D), are observed. For samples without pFBPA, the use of SiO_2-NPs provides no performance gain by itself, neither in terms of pFF nor R_s (Figure S13). Yet, for samples with pFBPA, the pFF is enhanced by up to 1.5%$_{abs}$ with the use of NPs, which appears to be due to an increased "shunt" resistance (R_{sh}) and reduced hotspot formation by the NPs (Figures 2F, 2G, and S14). As discussed in the previous section, the origin of the shunts is likely the large number of pinholes (with diameters of <100 nm) in our perovskite films due to pFBPA use, which is suppressed by the SiO_2-NPs, which might be reducing the probability of pinhole formation within a certain distance around them (Figure 2G). Notably, the pinhole density observed with the scanning electron microscope (SEM) is higher than the density of severe hotspots observed with lock-in thermography. This implies that not all the nanoscale pinholes result in shunts that are similarly detrimental for device operation but, rather, that the probability of having detrimental shunts is increased with the enhanced number of pinholes. Moreover, we note that the term "shunt" is used here to describe any defective location that passes more current than a location without such defects at the same operating voltage and is not necessarily of ohmic nature (i.e., with the current being directly proportional to the voltage, by a factor independent of voltage).[63,64] For instance, despite the clear increase in R_{sh} at low terminal voltages by the NPs, the R_{sh} is much higher than what would be effective on pFF and FF, even without the NP use (i.e., about 10^5 Ω.cm^2). Because ohmic shunts typically dominate the R_{sh} in this voltage regime, it can be inferred that the detrimental shunts in our devices are not of purely ohmic nature. Indeed, a non-ohmic current-voltage behavior can be present in pinholes with a direct ETL/HTL contact, with an impact in the operating voltage regime (i.e., about 1.0 to 1.2 V) that is relevant for the FF and pFF.[64]

In terms of R_s, a slight reduction (by about 0.5 Ω.cm^2, corresponding to an FF increase of roughly 0.7%$_{abs}$) is observed up to a NP concentration of 0.1 wt % despite the reduced contact area between the HTL and the perovskite absorber (Figure 2E). The change is larger with Me-4PACz, in which case the surface wettability is also influenced considerably with the NP concentration, which might be responsible for the more significant variations in the contact resistance between the Me-4PACz and the perovskite absorber at locations without the NPs. It is also possible that excess, unbound SAM molecules at the surface of the ITO are washed off by the ethanol in which the NPs are dispersed. Although there are reports that show washing has no significant influence on the device performance, it was also shown to be dependent on the perovskite composition.[24,65,66] Here, we checked the sole influence of ethanol washing using 2PACz (without the NPs) because fabricating SJ devices solely with Me-4PACz was not possible due to wettability issues. The results indicate minor changes with washing and without a clear conclusion on whether it improves or degrades the performance (Figure S15). Yet, we still acknowledge that the response of Me-4PACz to washing might be different than 2PACz and leave this as a possibility that needs further investigation. With the NP concentration increasing above 0.1 wt %, an increase in R_s is accompanied by an increasing degree of hysteresis for both SAMs under fixed voltage scanning conditions (Figure S16). Notably, the increase in R_s occurs more rapidly, and in some cases excessively, with 2PACz compared with Me-4PACz (Figure S17). This is possibly related to the higher—and in some cases higher than usual—contact resistance at the 2PACz/perovskite interface, whose effect on total device R_s is amplified with the higher NP concentrations (i.e., lower contact areas). For concentrations beyond 0.2 wt %, the increase in R_s and hysteresis (under constant scan speed) becomes very significant for both SAMs, making these configurations unsuitable for high-performance device operation. Whereas the increase in R_s with the NP coverage can be partially explained by the reduced perovskite/HTL contact area, the appearance of hysteresis indicates an additional change in the charge transport dynamics or the electric field in the absorber, e.g., by a variation in the electrostatics around the dielectric NPs, which can be significant in the case of densely packed configurations (e.g., >0.5 wt %). Notably, with the use of Me-4PACz/SiO_2-NPs, we also observe a sharper increase in R_s with the added pFBPA concentration as compared with the samples with bare 2PACz that were discussed in the previous section (Figure S3). This might be linked to the negative influence of pFBPA on the contact resistance at the bottom side with Me-4PACz (e.g., due to the increasing VBM offset with pFBPA), which is enhanced with a lower contact area fraction between the perovskite and the HTL due to the NP use. Overall, the results show that care must be taken when optimizing the SiO_2-NP concentration, which is affected by the choice of both the HTL and perovskite composition, both of which can influence charge transport at the HTL/perovskite interface.

In summary, our observations suggest a partial contact architecture in SiO_2-NPs similar to that of PERC cells, yet with some major differences. Similar to PERC cells, the configuration consists of (1) the insulating perovskite/SiO_2-NP interface with high-passivation quality and (2) the HTL/perovskite interface through which the current passes and with a passivation quality worse than that achieved with the SiO_2-NPs (for the HTLs tested in this work). However, because the non-radiative recombination rate is already quite low at the perovskite/HTL interface with 2PACz and Me-4PACz, and the recombination near the perovskite/C_{60} interface is much more limiting, the main source of PCE improvement by the NPs is not the improved passivation at the perovskite/HTL interface and the V_{oc}, such as in PERC cells. Notably, this finding is different from that of Peng et al., who report considerable improvements in non-radiative recombination due to the improved passivation by porous Al_2O_3 nanoplates, albeit using an absorber with a different band gap.[51] In our case, the pFF (and thus FF) improvement is driven by the reduction in the pinhole and shunt density (Figures 2F–2H) and a slight decrease in R_s (about 0.5 $\Omega.cm^2$) at low NP concentrations (<0.1 wt %). The replacement of 2PACz with Me-4PACz results in an additional decrease in R_s (by about 1.5 $\Omega.cm^2$) whereas the V_{oc} and the pFF remain similar. These findings are in alignment with those of Levine et al., which also observe improved hole transport properties and higher FFs with Me-4PACz as compared with 2PACz.[67] Notably, the lower R_s obtained with Me-4PACz contrasts with the findings of Siekmann et al., which show similar R_s for devices with 2PACz and Me-4PACz, yet it is possible that the differences in perovskite composition and use of SiO_2-NPs in our work are responsible for this discrepancy.[43]

Perovskite-Si tandem cells—design, performance, and reproducibility

To demonstrate the PCE potential of perovskite-Si tandem devices incorporating pFBPA and Me-4PACz/SiO_2-NPs, we integrate perovskite cells with front-side flat, rear-junction Si heterojunction (SHJ) bottom cells in a monolithic configuration

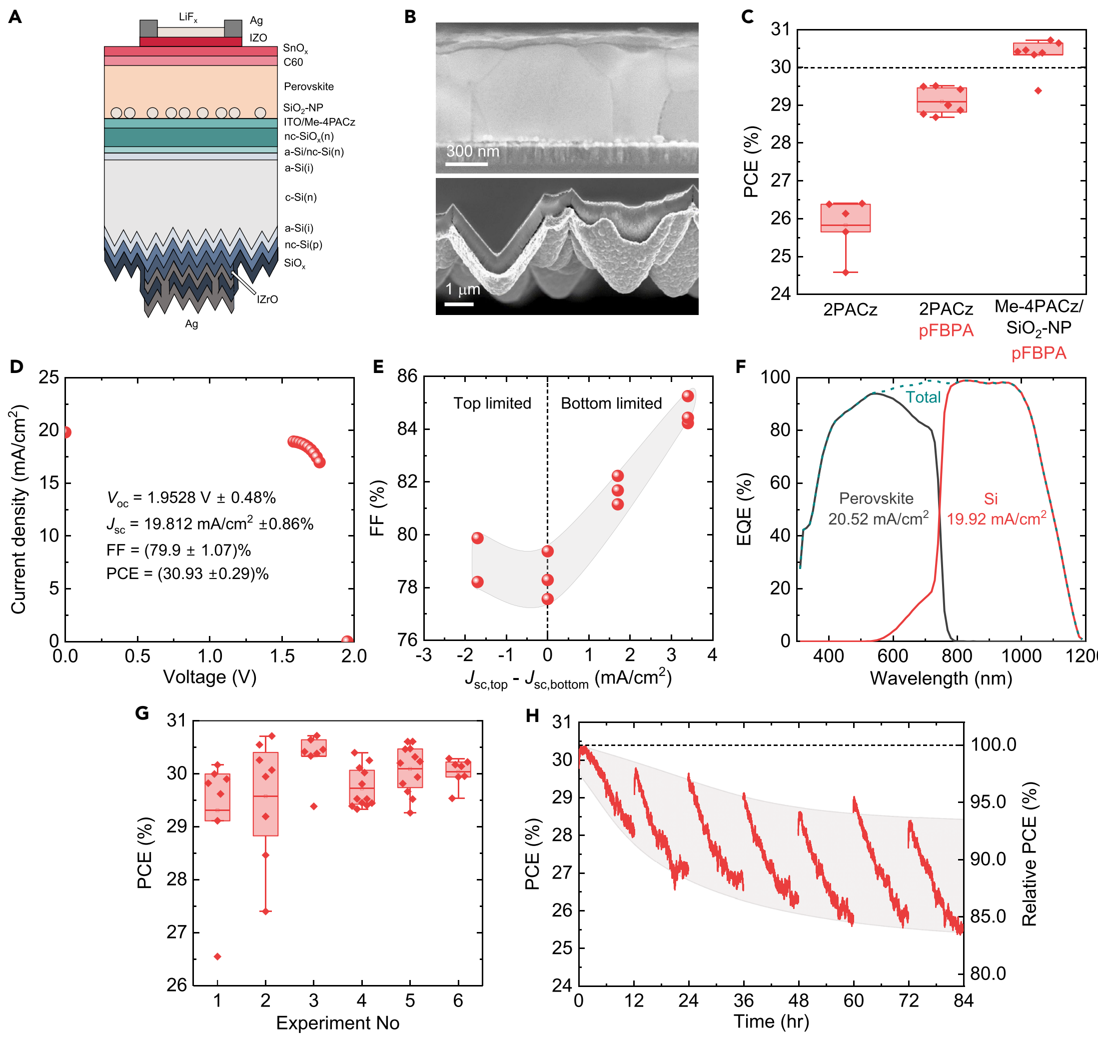

Figure 3. Overview of perovskite-Si tandem solar cells

(A and B) (A) Schematic representation and (B) SEM images of the cross-section view of the perovskite cell (top) and the metallized rear side of the Si cell (bottom) of a tandem solar cell.

(C) In-house-measured power conversion efficiencies (PCEs) for the highest performing batches (i.e., in terms of average PCE) fabricated with three different device configurations.

(D) Current density vs. voltage and the performance parameters of an independently certified 1.17-cm^2 perovskite-Si tandem solar cell.

(E) Fill factor (FF) vs. mismatch of short-circuit current densities of the perovskite ($J_{sc,top}$) and Si ($J_{sc,bottom}$) subcells.

(F) An exemplary external quantum efficiency (EQE) vs. wavelength spectrum for the perovskite and Si subcells measured in-house from a non-metallized location, with an intentionally imposed current mismatch to maximize the PCE.

(G) PCEs of 1.2 cm^2 perovskite-Si tandem solar cells with Me-4PACz/SiO_2-NP and 5 mM of pFBPA in the precursor fabricated in six consecutive batches. The data point marked by the star indicates the initially in-house-measured PCE for the independently certified cell.

(H) PCE of a solar cell measured by maximum power point (MPP) tracking with an artificially induced day-night cycle. The cell is measured for 12 h under 1-sun illumination and rested under dark conditions for 12 h, and the 24-h cycle is repeated for a week (i.e., a total of 84 h of operation). The "relative PCE" is the ratio of PCE at a given instant normalized to the maximum PCE of the cell recorded during this test (i.e., 30.4%).

(Figures 3A and 3B). The inverted structure, with the *p*-type contact at the bottom and *n*-type at the top, is common for most high-efficiency perovskite-Si devices of today due to the relatively transparent electron transport layers of the perovskite cell and the well-established SHJ technology that requires few modifications to be integrated in tandems. Overall, the use of pFBPA and Me-4PACz/SiO_2-NPs enables reproducible PCE gains of 3$\%_{abs}$ and 1$\%_{abs}$, respectively (Figure 3C). Ultimately, with the two developments combined, PCEs beyond 30% (reproducibly within

30 ± 1%) with an independently certified maximum of 30.9% were achieved, with exceptional shelf stability (Figures 3C, 3D, 3G, S18–S21, and S33). In the following, we outline some of the features of these devices and aspects that are specific to tandem device operation.

Our tandem devices have a simple optical design, without a low-refractive index rear-reflector, nanotextured Si surface, or light-management foil at the front side. Moreover, we use thinner Si wafers (about 190 μm thick) than those typically used in high-efficiency tandem devices (e.g., about 260–280 μm),[23,24,48,68] which leads to short-circuit current densities (J_{sc}) lowered by about 0.3 mA/cm^2 in the bottom cell. Nevertheless, the total J_{sc} of the subcells is roughly 40.5 mA/cm^2 in the non-metallized locations and 40 mA/cm^2 including the total metal grid shadowing of 1.5%. Two main contributors to these relatively high J_{sc} values, compared with the preceding record-efficiency devices,[24,48] are the use of a relatively thin front transparent conductive oxide (of roughly 35 nm, 180 Ω/sq) to minimize the parasitic light absorption and the use of a fine Ag grid (i.e., 60-μm-wide fingers, Figure S22) to minimize the lateral transport losses in the TCO at the front side. To maximize the PCE, the total J_{sc} should be divided into the subcells based on their individual electrical performances (e.g., pFFs).[69–71] In our case, the PCE is maximized not by current-matching the subcells but by tuning the band gap and the thickness of the top cell to generate roughly 0.5 mA/cm^2 more J_{sc} compared with the bottom cell (i.e., bottom-limited devices), as estimated by controlled current-mismatch measurements and numerical calculations based on an equivalent-circuit model (Figures 3E, 3F, and S23). The asymmetry requirement is due to the relatively poor electrical performance of the 1.2-cm^2 top cell, with an estimated pFF of 84% to 85% that is 3%$_{abs}$ to 4%$_{abs}$ lower than what we achieve with opaque 0.1 cm^2 devices on ITO-coated glass substrates (e.g., Figure 1D). The difference in top cell performances between these two configurations is yet to be fully clarified. However, a performance decrease with area up-scaling is common for perovskite devices, e.g., due to the difficulty in maintaining high film quality over a large area.[10,11] The different substrates used in tandem and SJ devices (e.g., in-house-deposited interconnection ITO on Si vs. commercial glass/ITO) can also be responsible for the performance difference.[72]

In terms of the bottom cell performance, the 1.1-cm^2 Si devices exhibit a pFF of around 83% to 84%. Besides the decrease of about 1%$_{abs}$ compared with a 4-cm^2 cell, which is expected due to enhanced perimeter recombination losses, a notable problem due to area scaling is not observed (Figure S24). Nevertheless, we highlight here a simple but critical feature of our Si cells that helps preserve their high performance through the numerous top cell fabrication steps in which the rear side gets in contact with various surfaces (e.g., the hot plate). Particularly with the *pn* junction situated at the rear side, the cells are highly sensitive to mechanical damage, and even a small scratch in the metallized active area disrupting the thin-film Si layers can be detrimental to the pFF and FF. Damage to the non-metallized perimeter area, which is even more susceptible to external effects due to the lack of a TCO/Ag stack, can also degrade the performance, although to a lesser extent (Figure S25). Thus, to minimize the damage on the rear side of our Si cells, we deposit a full-area, 500-nm-thick SiO_x by plasma-enhanced chemical vapor deposition (PECVD) after the 150-nm-thick rear Ag deposition. To electrically contact the cells through the SiO_x film, we deposit an additional layer of Ag on the active area, which results in an Ag/SiO_x/Ag stack with a total thickness of about 800 nm. We speculate that Ag/Ag contacts form locally through the pinholes in the presumably imperfect SiO_x film, enabling a low contact resistance between the two Ag sheets without additional effort to realize well-defined contact openings. This configuration enables protection for both the active area and the non-metallized perimeter, resulting in a considerably enhanced FF, PCE, and reproducibility for the tandem devices (Figures S26 and S27). Although in extreme cases in which the cells were improperly handled by the experimenter (i.e., without attention to mechanical damage on the rear side), PCE gains of up to $10\%_{abs}$ on average were observed using the protection layer, while gains of about $1\%_{abs}$ were also recorded in less severe cases. These gains in PCE, which are larger than what can be achieved with most controlled modifications to both subcells, show the importance of a rear protection scheme for these small-area devices.

In terms of the tandem V_{oc}, the typically measured values range between 1,880 and 1,920 mV (Figure S20B). Whereas the Si bottom cell has a contribution of 700 to 710 mV to the tandem V_{oc}, the estimated contribution of the perovskite top cell is around 1,180 to 1,210 mV, which is slightly higher than the V_{oc} we typically measure with our SJ devices (i.e., 1,160 to 1,190 mV, Figures S3, S11, S12, S15, and S16). The reason for the difference between the V_{oc} of perovskite cells in SJ and tandem configurations is mainly the increased ratio of the area of the illuminated and dark regions of the devices (i.e., resulting in a smaller dark current density for the tandem design), which results in about a 20-mV difference between the two configurations (Figures S28 and S29). We also note that the outstanding V_{oc} of our certified device (i.e., 1,953 mV) can be considered an outlier and is also likely influenced by the light-soaking-induced degradation-recovery cycles following in-house measurements (which were rated at 1,935 mV in the latest measurements), which can result in an increase in the V_{oc} by about 20 mV (Figures S1 and S33).

Finally, the operational stability was also tested by MPP-tracking a device under 1-sun illumination (without a UV filter), ambient air (25ºC, 30% to 50% humidity), without encapsulation. Following a 12-h-long operation, the cell was stored in the dark for 12 h in N_2 ambient conditions and the cycle was repeated to imitate conditions similar to the day-night cycle that naturally occurs within a day, i.e., ISOS-LC-1 conditions. As shown in Figure 3H, after 12 h of continuous illumination, the PCE reduces to about $90\%_{rel}$ of the maximum PCE recorded within the day, yet it recovers to a large extent after storage under dark (by $5\%_{rel}$ to $10\%_{rel}$) conditions. At the end of a week-long test (i.e., 84 h of operation), the device still operated between 85% and 95% of the maximum PCE. Overall, two PCE loss mechanisms seem to be prevalent—a faster and recoverable one that occurs within a day, which can be related to, e.g., the migration of mobile ions within the device (that can also be responsible for the increased level of hysteresis at the end of one cycle, Figure S35), and a slower and possibly permanent one, which can be related to changes in the atomic composition of various layers of the device due to both intrinsic (e.g., light and electric bias) and extrinsic (e.g., ambient air) factors.[73–75] Although improving the intrinsic stability requires modifications to the composition of the perovskite or the fabrication methods, extrinsic stability can be improved by, e.g., encapsulating the devices. We emphasize, however, that the devices presented here are designed to have maximized PCE without specifically optimizing for long-term stability. Although recent results suggest that improved efficiencies correlate with improved stability under extended light-soaking for perovskite SJ devices, further research is needed for optimization and accurate evaluation of tandem devices in such aspects, which can notably also differ from SJ devices due to added complexity of electronics in a tandem device.[76,77]

DISCUSSION

Perovskite-Si tandem solar cell technology is the most prominent contender for the next generation of commercial solar cells. Yet, high PCEs first need to be realized to enable a cost-effective deployment of the technology, and the identification of high-performing device configurations is vital in this regard. Here, we outlined some of the main features of our front-side flat tandem devices and aspects that are crucial for high performance and reproducibility. Although the tandem devices based on front-side flat Si wafers are limited in optical performance compared with those based on textured ones, their PCE potential is of considerable interest as they are fully compatible with solution processing and require little change in perovskite processing conditions that are commonly used for SJ devices. Overall, the pFBPA use in the precursor enables significant gains in PCE through a reduction of interfacial recombination and R_s, and the approach is particularly attractive for industrialization of the technology as it does not require an additional deposition step for a passivating layer. The gain that can be achieved with this additive can be further enhanced and maximized if used in combination with a SiO_2-NP-coated substrate, which prevents the formation of pinholes and shunts due to the pFBPA use. The wettability-enhancing SiO_2-NPs also enable the reliable use of high-performance Me-4PACz as an HTL that further reduces the R_s and improves in particular the FF of the devices. Integrating these two main developments into an optically and electrically optimized tandem cell (e.g., with an optimized front TCO/Ag grid and a durable bottom cell), reproducible PCEs in excess of 30% (i.e., certified 30.9%) are achieved.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Christian M. Wolff (christian.wolff@epfl.ch).

Materials availability

This study did not generate new, unique materials.

Data and code availability

The datasets generated during this study are available from the lead contact upon reasonable request.

Fabrication of single-junction perovskite cells

The triple-cation perovskite (TCP, $Cs_{0.05}[FA_{0.90}MA_{0.10}]_{0.95}Pb[I_{0.80}Br_{0.203})$ absorber precursor was prepared using four stock solutions of (1) 1.55 M FAI (Dyenamo, >99.99%) and 1.7 M PbI_2 (TCI, >99.99%) in DMF:DMSO = 4:1; (2) 0.775 M FABr (Dyenamo, >99.99%), 0.775 M MABr (Dyenamo, >99.99%), and 1.7 M $PbBr_2$ (TCI, >99.99%) in DMF:DMSO = 4:1; (3) 1.5 M CsI (Alfa Aesar, >99.9%) in DMSO; and (4) pFBPA (Sigma Aldrich, 97%) in 28 mg/mL in DMF:DMSO = 4:1. Prior to the depositions, the solutions were mixed with a volume ratio of a:b:c:d = 80:20:5:X, respectively, to obtain the TCP precursor. The 2PACz (TCI) and Me-4PACz (Sikemia, >97%) were dissolved in ethanol (1 mg/mL). For the SiO_2-NP dispersion, synthetic amorphous colloidal silica was used. A 1 wt % stock suspension was prepared in ethanol and further diluted by ethanol to obtain concentrations ranging from 0.05 to 0.5 wt %. The particle size distribution was controlled by means of dynamic light scattering (DLS, Zetasizer nano, Malvern Panalytical) and an average particle size of 24 nm was measured. For device fabrication, commercial 2.5 × 2.5 cm^2 glass/ITO (Kintec, 15 Ω/sq) substrates were cleaned by Hellmanex in an ultrasonic bath for 15 min, followed by a deionized water rinsing for another 15 min. The samples were then dried by N_2 and received a UV-ozone treatment for 15 min. The samples were then transferred to a N_2-filled glovebox and a SAM HTL (i.e., 2PACz or Me-4PACz) was spin coated with a multi-step recipe—first, the SAM solution was dropped on the sample and rested for 10 s, next, the sample was spun for 30 s and, 15 s into the recipe, 100 μL of SAM was again dropped on the sample. The sample is then dried on a hotplate at 100°C for 10 min. The SiO_2-NP was coated statically by spin coating at 2,000 rpm and drying on a hotplate at 100°C for 10 min. The perovskite precursor ink was dropped on the substrate prior to spinning and 400 μL of anisole was dropped 20 s after the start of the spin at 3,500 rpm, which lasted for a total of 35 s. The samples were then annealed on a hotplate at 100°C for 15 min. A 20-nm-thick C_{60} layer (NanoC, >99.95%) was deposited by thermal evaporation. A 25-nm-thick SnO_x layer was deposited by atomic layer deposition at a temperature of 100°C. Finally, a 130-nm-thick Ag layer was deposited through a shadow mask by thermal evaporation to realize 3 cells (0.25 cm^2 metallized, 0.1 cm^2 illuminated area) per substrate.

Fabrication of perovskite-Si tandem cells

The rear-junction Si bottom cells were fabricated using a roughly 190-μm-thick, 2 Ω.cm, n-type, float-zone, shiny-etched monocrystalline Si wafers. After covering the front side of the wafers with a plasma enhanced chemical vapor deposition (PECVD)-grown SiN_x layer, a random pyramid texture was formed at the rear side using a potassium hydroxide solution. The SiN_x at the front side was removed after the texturization, and the wafers received a wet-cleaning procedure. Prior to the PECVD processes, the wafers were dipped in a hydrofluoric acid solution to remove the thin, chemical oxide at the surface. All hydrogenated amorphous and nanocrystalline Si layers were deposited by a PECVD system, with a plasma frequency of 13.56 MHz. On the front side, hydrogenated intrinsic and phosphorus-doped amorphous Si layers, [a-Si:H(i) and a-Si:H(n)], and the nanocrystalline phosphorus doped Si and SiO_x [nc-Si:H(n) and nc-SiO_x:H(n)] layers were deposited at 200°C. On the rear side, the a-Si:H(i) deposition at 200°C was followed by the depositions of ultra-thin SiO_x and nc-Si:H(p) layers at 175°C. A zirconium-doped indium oxide (IZrO) and Ag stack was deposited through a shadow mask, resulting in a contact area (1.3 × 1.3 cm^2) slightly larger than the active area of the completed tandem cell (1.1 × 1.1 cm^2). A full-area 500-nm-thick SiO_x layer was deposited on the rear side, covering both the metallized and non-metallized regions. To finalize the rear contact, an additional layer of Ag was sputtered through a shadow mask on the active area. The front side was finalized by sputtering a full-area, roughly 20-nm-thick ITO layer. The 4″ wafers were then laser cut down to 2.5 × 2.5 cm^2 substrates and annealed at 210°C for 30 min in an oven operating in ambient air. The perovskite cells were deposited on top following the same procedure for SJ cells with some differences—instead of a 1.5 M solution, a 1.7 M solution was used to obtain thicker films that satisfy the desired current-matching conditions. A 15-nm-thick C_{60} layer was used, instead of 20 nm used in SJ devices. Following the SnO_x depositions, a 35-nm-thick IZO layer (around 180 Ω/sq) was deposited by RF sputtering with Ar through a stainless-steel shadow mask. A 700-nm-thick Ag layer was deposited by thermal evaporation through a stainless-steel shadow mask with 60-μm-wide fingers (with 3.6 mm spacing in between subsequent fingers) and a 750-μm-wide busbar that surrounds the active area. To protect the samples from damaging by needle probes in subsequent device measurements, a low-temperature Ag paste was applied to a pad area (of about 1 mm^2) by a handheld brush, and the samples were annealed at 70°C for 10 min on a hotplate under ambient air. Finally, a 100-nm-thick LiF_x layer was full-area deposited on the samples by thermal evaporation.

Current-voltage measurements of the solar cells

The current-voltage measurements under simulated AM 1.5G illumination were conducted using a Wacom solar simulator (AAA) equipped with Xe and Ha lamps. For SJ perovskite devices, each pixel was measured individually while masking the other two pixels (out of three) on the substrate with a shadow mask to eliminate unintended light-soaking. A forward (i.e., near short circuit to open circuit) and reverse (i.e., near short circuit to open circuit) scan was conducted for each device within 12 h following the completion of the device fabrication. The terminal voltage was varied between −0.1 V and 1.3 (2) V for SJ (tandem) devices at a scan speed of roughly 200 (350) mV/s. For brevity, the results of only the reverse scans are presented throughout the manuscript. For devices without pFBPA, a preliminary light-soak pretreatment at open-circuit for 3 min was applied to maximize their performance, in accordance with the experimental results in Figure S1. Devices with pFBPA did not undergo any pretreatment unless specified otherwise. A temperature control was not used in SJ perovskite cell measurements. For Si and tandem device measurements, a temperature-controlled (25ºC) brass chuck was used. The tandem measurements were conducted with a Kelvin needle probe at the top side and with an isolated pin integrated in the brass chuck at the rear side, with a 4-wire configuration on each side. A rubber shadow mask was used to illuminate solely the active area on the Si substrates (1.2 cm^2). The long-term MPP measurements of the tandems, which lasted 48 to 84 h, were performed out of the regular testing scheme and included 9 to 12 months old samples. The current-mismatch measurements were performed by selectively exciting the subcells by light emitting diodes with peak wavelengths of 428 and 950 nm. The pFF is the FF of the pseudo-JV curve (i.e., the idealized curve free of R_s) obtained by measuring the V_{oc} of the cells at two additional light intensities (0.05 and 0.1 sun) to extract the ideality factor near the MPP conditions and to construct the current vs. voltage curve from the relation $V_p = V_{oc}$ (sun) = $J_{sc} \times (1 - \text{sun})$, as described in previously in literature.[78] The R_s was calculated by dividing the voltage difference between the actual (V_{mpp}) and pseudo-JV curves ($V_{p,mpp}$) by the current density at MPP of the actual curve ($R_s = [V_{mpp} - V_{p,mpp}]/J_{mpp}$).[79,80]

External quantum efficiency measurements

Measurements were conducted with a custom-built spectral response setup where the samples are irradiated with chopped monochromatic light at a frequency of 217 Hz, with a spot size of about 1 mm^2, and the response was measured with a lock-in amplifier. For tandem cells, white and blue LEDs were used to ensure that the subcell being measured was the current limiting one. To measure each subcell near short-circuit conditions, 0.7 and 1.2 V bias voltages were applied to the cells when measuring the top and bottom cells, respectively.

Lock-in thermography measurements

Images were captured in a Thermosensorik system using AC bias alternating between 0 and 1 V at 25 Hz.

Scanning electron microscope imaging

Images were acquired with acceleration voltages ranging from 1 to 5 kV using an in-lens detector (Zeiss NVision 40).

XPS measurements

The measurements were performed using Axis Supra (Kratos Analytical) with a monochromatic Al Kα X-ray line. The pass energy was set to 20 eV with a step size of 0.1 eV. The samples were electrically grounded to limit charging effects.

XRD measurements

XRD patterns were obtained using an Empyrean diffractometer (Panalytical) equipped with a PIXcel-1D detector with Cu Kα radiation.

Steady-state PLQY measurements

Custom-built steady state PL setup is utilized to measure photoluminescence (PL) and PL quantum yield (PLQY). The light from the laser diode with a peak wavelength of 532 nm is coupled into a fiber directed into the integrating sphere and sample is illuminated from the glass side of the samples. Then the emission from the sample is homogenized by multiple reflections within the integrating sphere and coupled out into another fiber that is connected to a spectrometer. For calibration, three fluorescent test samples with a PLQY of around 70%, supplied from Hamamatsu Photonics, were measured to obtain a relative error of less than 5%. The quasi-Fermi-level splitting loss due to non-radiative recombination at 1-sun illumination ($qV_{\text{non-rad}}$) was calculated by the product of kT and ln(PLQY).[81,82]

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2024.04.015.

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AUTHOR CONTRIBUTIONS

Conceptualization, D.T., K.A., X.-Y.C., S.-J.M., A.W., G.A., N.B., M.B., Q.J., and C.M.W.; methodology, D.T., D.A.J., and C.M.W.; validation, D.T.; investigation, D.T., K.A., D.A.J., M.M., H.L., F.F., and C.M.W.; writing – original draft, D.T.; writing – review & editing, D.T., K.A., and C.M.W.; visualization, D.T.; supervision, M.B., Q.J., C.M.W., and C.B.; project administration, M.B., Q.J., C.M.W., and C.B.; funding acquisition, M.B., Q.J., C.M.W., and C.B.

DECLARATION OF INTERESTS

X.-Y.C. and Q.J. have filed a patent related to the usage of pFBPA for perovskite photovoltaics. S.-J.M. and Q.J. have filed a patent related to the usage of SiO_2 nanoparticles for perovskite photovoltaics.

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